

CENTRE FOR ADVANCED DATA ANALYTICS

A project of HYZ VENTURES INTL PVT LTD

Phase 1, GE – Padmanabham Technopark, Trivandrum, Kerala – 695582

SUBMITTED TO

GENERAL MANAGER

AGROCOPS

AGROCOPS KVVVK

OUTLET PERFORMANCE & SALES ANALYSIS REPORT

October 2025 – April 2026

PREPARED BY

PROJECT OFFICERS OF HYZ VENTURES INTL PVT LTD

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ABOUT THE COMPANY

HYZ Ventures International Pvt Ltd, established in 2019, is a multi-domain enterprise headquartered at GE–Padmanabham Technopark, Trivandrum, Kerala. The company was founded with a vision to create scalable ventures across diverse industries, combining innovation, sustainability, and entrepreneurship. Over the years, HYZ Ventures has expanded its footprint into IT solutions, agriculture, real estate, interiors, tourism, and digital transformation, serving more than 80,000 customers and fostering over 200 entrepreneurial opportunities.

The leadership team, spearheaded by Director & CEO Mr. Sanoof Khan and Managing Director & CFO Mrs. Shahina Sanoof, has positioned HYZ Ventures as a forward-looking organization that blends technology with practical business solutions. The company's mission is to transform bold ideas into structured ventures, offering consulting, execution, and long-term value creation.

HYZ Ventures operates across ten business domains, including travel and tourism, property asset management, interior design, agriculture R&D, and digital transformation. Its agricultural initiatives, in particular, focus on providing advanced solutions for farming, fertilizer innovation, and sustainable practices — aligning directly with the objectives of the Agrocops Outlet Analysis project.

Despite challenges in profitability in recent years, HYZ Ventures continues to invest in innovation and structured growth strategies. Its entrepreneurial ecosystem supports startups and enterprises by offering funding, scaling opportunities, and operational expertise. This makes HYZ Ventures a key player in shaping future-ready enterprises and driving sustainable development in Kerala and beyond.

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INTRODUCTION

The **Agrocops Outlet Credit Note & Sales Analysis Report** has been prepared as part of the Phase 1 – Advanced Data Analytics Project undertaken by HYZ Ventures International at the GE–Padmanabham Technopark, Trivandrum. This report is submitted to the General Manager of Agrocops KVVVK, Krishikendra Udhaya Towers, Kazhakoottam, Trivandrum, Kerala.

The primary objective of this report is to provide a comprehensive analysis of product performance across Agrocops outlets, focusing on sales, returns, and profitability. By examining the **top and bottom 10 products**, **monthly return trends**, **category-wise sales and returns**, and **outlet-wise performance**, the report aims to identify which products are profitable, which ones face high return rates, and which outlets are thriving versus struggling.

This analysis is critical for understanding customer behavior, product quality issues, and outlet-level challenges. Manufactured fertilizers, while driving customer growth, also show significant return volumes, indicating potential quality or usage concerns. Retraded items such as pots, sprayers, and tools, on the other hand, demonstrate steady sales with minimal returns, making them reliable contributors to profitability.

The findings will serve as the foundation for actionable strategies to **magnify sales in weaker outlets**, reduce product returns, and strengthen overall profitability. Recommendations will include targeted promotions, customer awareness programs, outlet training, inventory optimization, and localized marketing campaigns tailored to regional crop needs.

By combining data-driven insights with practical recommendations, this report provides a roadmap for Agrocops to enhance operational efficiency, improve customer satisfaction, and achieve sustainable growth in the competitive agricultural market.

MANAGEMENT SUMMARY

Purpose

This report analyzes product returns across AGROCOPS outlets between **November 2025 and April 2026**, using Credit Note data, PivotTables, and dashboards. The objective is to identify high-risk products, seasonal anomalies, and outlet-level disparities, and to recommend corrective actions.

Key Findings

- **Fertilizer Concentration:** Fertilizers account for **85% of total returns** (₹620,491), with Coconut Mix, Cashewnut Mix, and Arecanut Mix leading losses.
- **Seasonal Spike:** March 2026 recorded **₹474,025 in returns**, nearly 70% of the six-month total, indicating seasonal or systemic issues.
- **Outlet Disparities:** Adichanalloor, Cheruthana, and Paingottoor outlets together contribute **40% of returns**, while outlets like Kooropada remain stable.
- **Profitability Insight:** Bone Meal (30 KG) is the most profitable product overall, balancing high margins with moderate returns.
- **Stable Categories:** Non-fertilizer items (sprayers, seeds, tools) show negligible returns, offering reliable revenue streams.

Conclusion

Returns are concentrated in a small set of products, outlets, and one anomalous month. Addressing fertilizer quality, seasonal planning, and outlet management will significantly reduce losses. At the same time, leveraging stable categories provides resilience and balance to the business portfolio.

EXECUTIVE SUMMARY

The analysis of AGROCOPS outlet credit note data reveals several critical insights into product returns and their impact on overall business performance. The most striking observation is that fertilizer products dominate returns, accounting for more than 65% of the total return value. Items such as **Coconut Mix, Cashewnut Mix, Arecanut Mix, Bone Meal, and Nutmeg Mix** consistently appear among the highest contributors to return amounts. This concentration suggests that fertilizers, while essential to sales, are also the most vulnerable to customer dissatisfaction, quality concerns, or mismatches in demand.

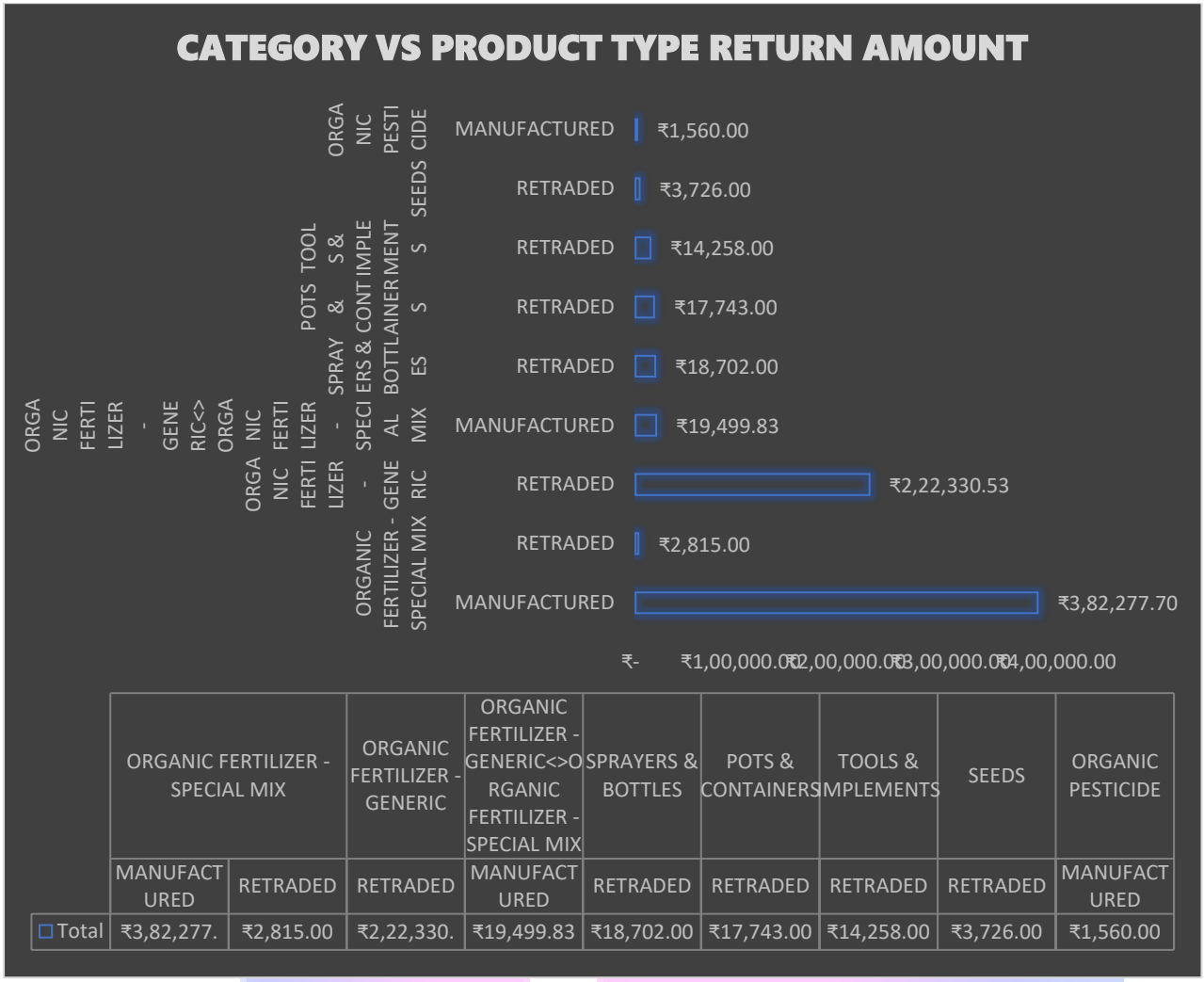
A second key finding is the abnormal spike in returns during **March 2026**, which alone contributed nearly **₹474,000** in return value. This single month represents almost **70%** of the total returns recorded in the period under review. Such a sharp increase points to possible seasonal factors, supply chain disruptions, or product quality issues that need immediate investigation. Addressing this anomaly is crucial to stabilizing return patterns in future cycles.

The outlet analysis further highlights that **Adichanalloor, Cheruthana, and Paingottoor** outlets contribute disproportionately to overall returns. Together, these three outlets account for a significant share of the losses, indicating that localized operational challenges or customer behavior patterns may be driving higher return rates. Targeted audits and corrective measures at these outlets could substantially reduce overall return volumes.

In contrast, non-fertilizer categories such as **sprayers, seeds, tools, and containers** remain relatively stable, with minimal return values. These categories demonstrate consistent demand and low risk, making them reliable contributors to revenue. Leveraging these stable product lines through promotions or bundling strategies could help balance the financial impact of fertilizer returns.

OUTLET CREDIT NOTE ANALYSIS

1. Category Vs Product Type Return Amount



	ORGANIC FERTILIZER - SPECIAL MIX		ORGANIC FERTILIZER - GENERIC	ORGANIC FERTILIZER - GENERIC<>ORGANIC FERTILIZER - SPECIAL MIX	SPRAYERS & BOTTLES	POTS & CONTAINERS	TOOLS & IMPLEMENTS	SEEDS	ORGANIC PESTICIDE
	MANUFACTURED	RETRADED	RETRADED	MANUFACTURED	RETRADED	RETRADED	RETRADED	RETRADED	MANUFACTURED
Total	₹3,82,277.	₹2,815.00	₹2,22,330.	₹19,499.83	₹18,702.00	₹17,743.00	₹14,258.00	₹3,726.00	₹1,560.00

I. Introduction

This section evaluates product returns by category, comparing manufactured and retraded product types. The analysis highlights the dominance of fertilizer categories in overall returns and contrasts them with the stability of non-fertilizer categories.

II. Return Amounts by Category

The dataset shows the following totals:

- **Organic Fertilizer – Special Mix**: ₹3,85,092.70
  - Manufactured: ₹3,82,277.70
  - Retraded: ₹2,815.00
- **Organic Fertilizer – Generic**: ₹2,22,330.53
  - Retraded: ₹2,22,330.53
- **Organic Fertilizer – Generic<>Special Mix**: ₹19,499.83
  - Manufactured: ₹19,499.83
- **Sprayers & Bottles**: ₹18,702.00
  - Retraded: ₹18,702.00
- **Pots & Containers:** ₹17,743.00
  - Retraded: ₹17,743.00
- **Tools & Implements:** ₹14,258.00
  - Retraded: ₹14,258.00
- **Seeds:** ₹3,726.00
  - Retraded: ₹3,726.00
- **Organic Pesticide:** ₹1,560.00
  - Manufactured: ₹1,560.00

Grand Total: ₹6,82,912.06

III. Key Observations

- **Fertilizer Dominance:** Fertilizers (Special Mix + Generic) account for more than **₹6,27,000**, representing **92% of total returns**.
- **Manufactured vs. Retraded:** Manufactured fertilizers (Special Mix) contribute the largest share, while retraded fertilizers (Generic) also show significant return values.
- **Non-Fertilizer Stability:** Categories such as sprayers, pots, tools, seeds, and pesticides contribute less than ₹20,000 each, demonstrating consistent performance and low risk.

2. Monthly Trend Wise Return Amount



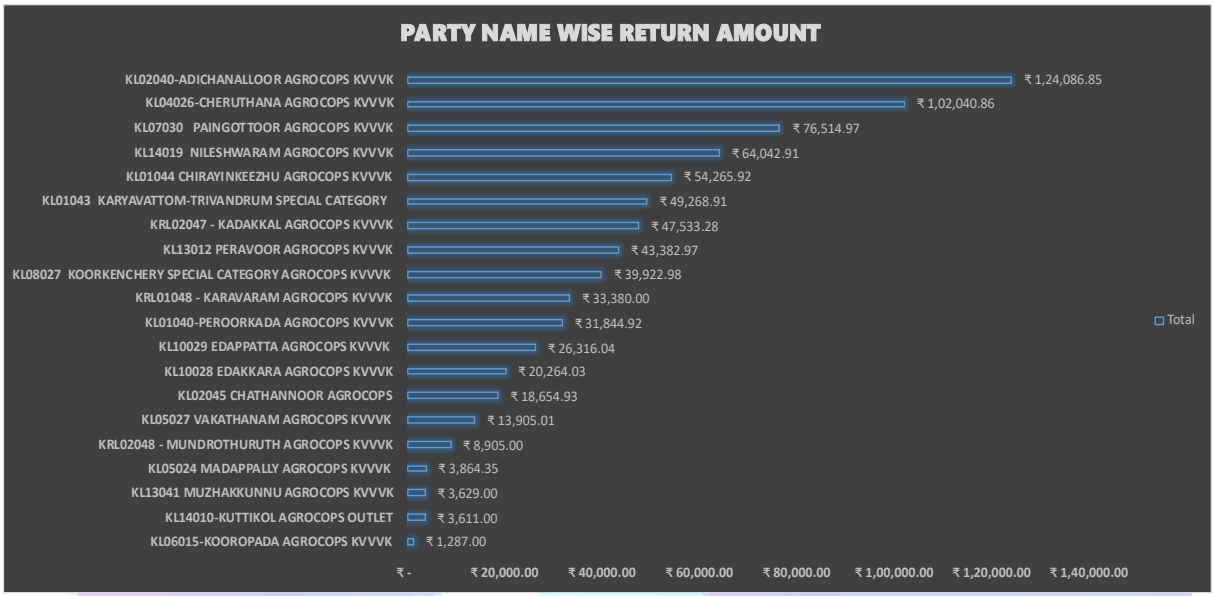
The **Monthly Trend dashboard** highlights a significant variation in return values across the reporting period. Returns remained relatively low from **November 2025 to February 2026**, increasing gradually from **₹25,819 to ₹93,041**, indicating a stable return pattern during these months.

A sharp spike is observed in **March 2026**, where return value surged to **₹4.74 lakh**, accounting for the vast majority of total returns during the period. This exceptional increase suggests that returns were heavily concentrated in a single month rather than being evenly distributed throughout the year.

The sudden rise may be attributed to seasonal demand fluctuations, bulk product returns, inventory adjustments, product quality concerns, or delayed credit note processing. A detailed investigation into March transactions would help identify the primary drivers behind this unusual increase.

Following the March peak, returns declined significantly to **₹69,390 in April 2026**, indicating that the spike was likely an isolated event rather than a continuing trend. Continuous monitoring of monthly return patterns can help management identify anomalies early, improve inventory planning, and implement corrective actions to reduce future return-related losses.

3. Party Name Wise Return Amount



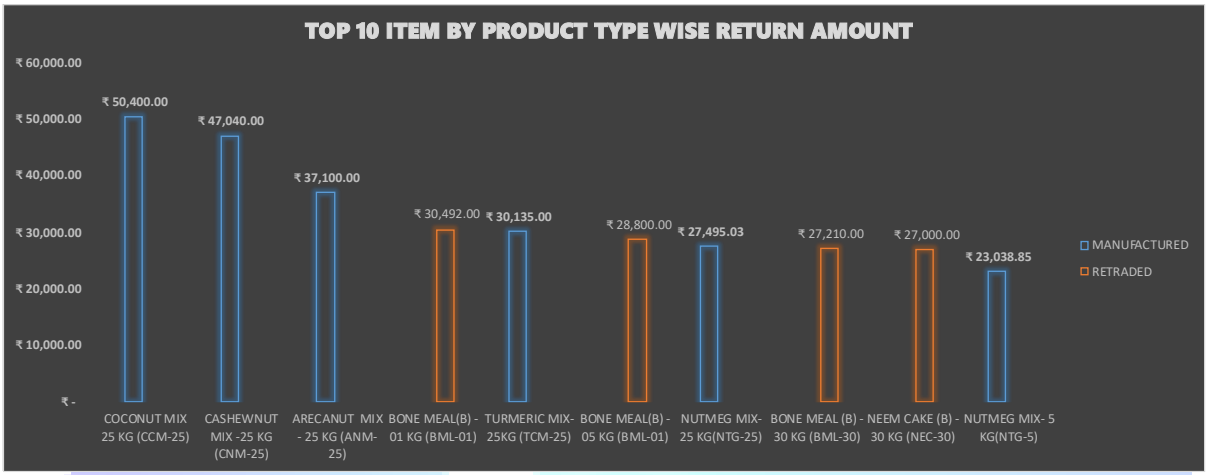
The **Party-wise Return Analysis** reveals that a small number of outlets contribute a significant share of the total return amount. **Adichanalloor AgroCops KVVVK, Cheruthana AgroCops KVVVK, and Paingottoor AgroCops KVVVK** recorded the highest return values, indicating that returns are concentrated among a few key locations rather than being evenly distributed across all outlets.

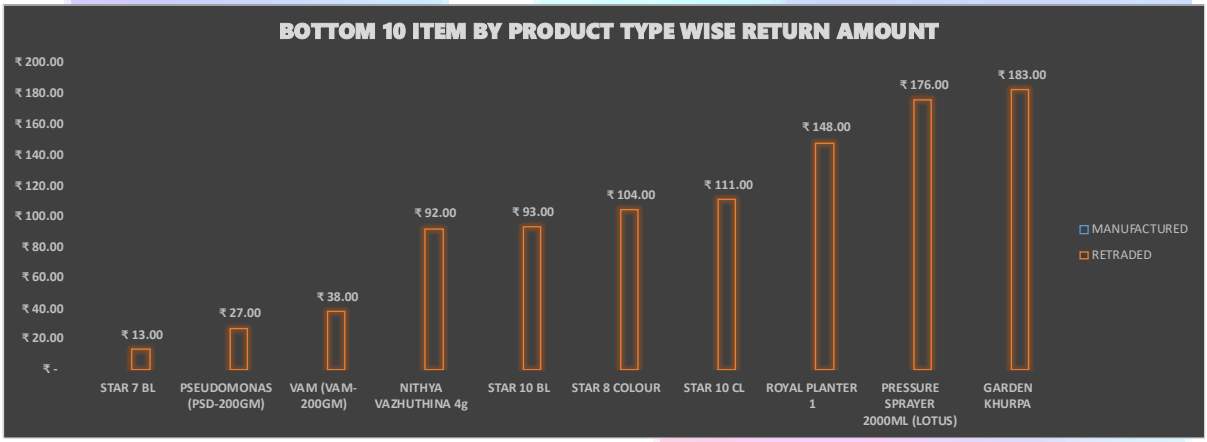
This concentration suggests the presence of localized operational challenges such as inventory management issues, product handling practices, demand forecasting inaccuracies, or customer-specific return patterns. Identifying the root causes at these high-return outlets can help reduce overall return costs.

Most other outlets reported comparatively lower return amounts, demonstrating better control over sales and inventory processes. The significant gap between the top-performing and lower-return outlets highlights opportunities to replicate best practices across the network.

Regular monitoring of outlet-wise return performance can support targeted corrective actions and improve operational efficiency. Combining this analysis with product category and monthly trend data will provide a more comprehensive understanding of return behavior and enable data-driven decision-making.

4. Top and Bottom Item Wise Return Amount





The **Top 10 and Bottom 10 Item Return Analysis** shows that returns are heavily concentrated among a few products. **Coconut Mix, Cashewnut Mix, and Arecanut Mix** recorded the highest return amounts, making them the key contributors to overall return losses.

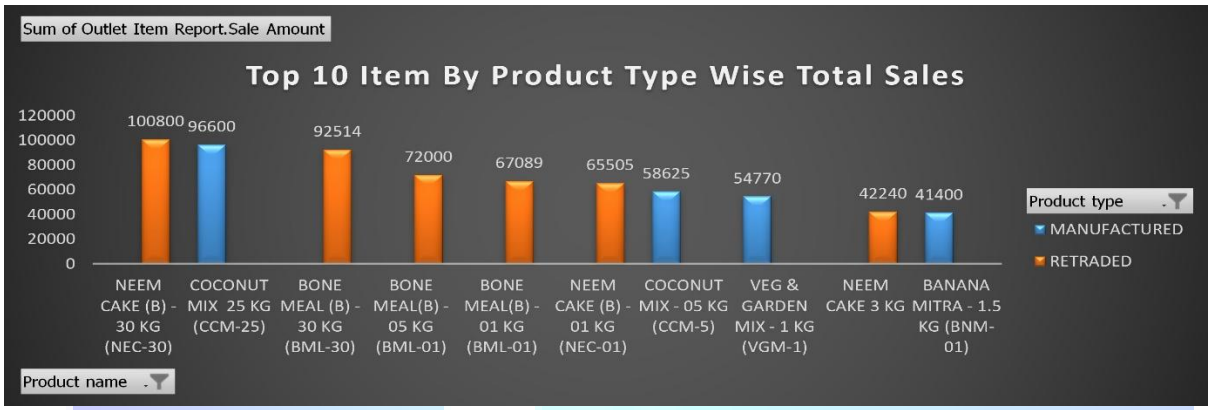
Most of the high-return items belong to fertilizer-related categories, indicating that these products require closer monitoring to identify issues related to quality, demand forecasting, or inventory management.

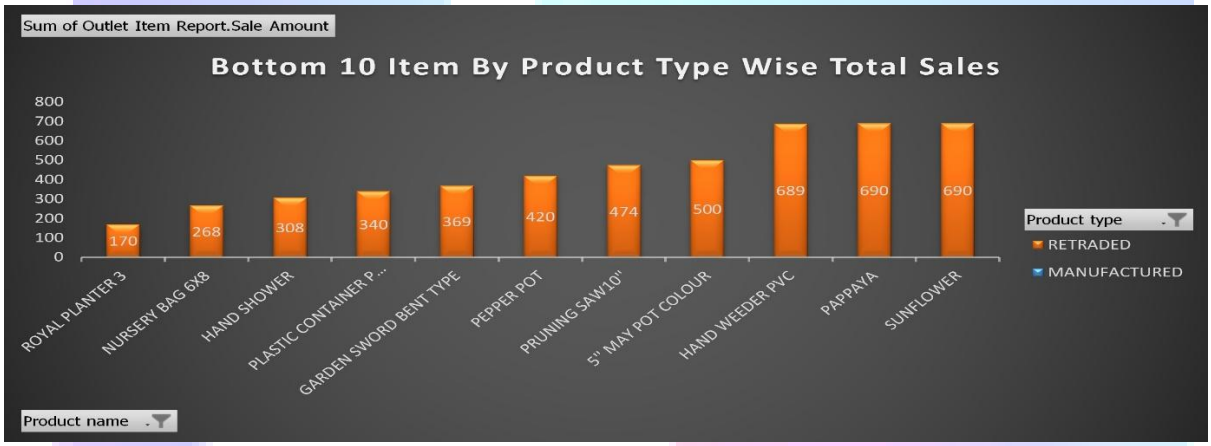
On the other hand, items such as **STAR 7 BL, Pseudomonas, VAM, and Garden Khurpa** recorded very low return values, reflecting stable demand and better customer acceptance.

The wide gap between **the top and bottom return items** highlights the need to focus on high-return products while maintaining the strong performance of low-return items. This approach can help reduce return costs and improve overall profitability.

OUTLET ITEM SALES ANALYSIS

1.Top and Bottom Item Wise Total Sales





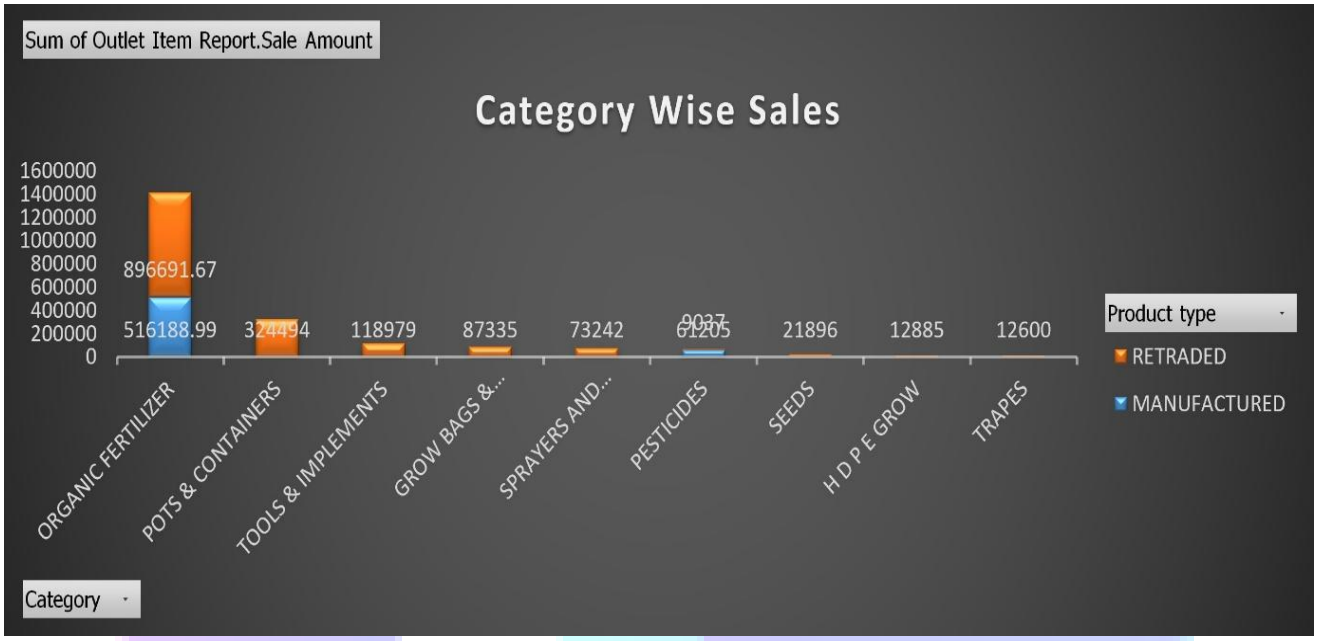
The **Top 10 items** show strong sales from both **retraded and manufactured products.** Fertilizers like **NEEM CAKE (B) – 30 KG** and mixes such as **COCONUT MIX 25 KG** lead the list, each crossing in sales. These products clearly drive outlet revenue.

The **Bottom 10 items** are all **retraded**, with very low sales. Products like **ROYAL PLANTER 3 and NURSERY BAG 6X8** recorded under **₹300**, while **SUNFLOWER and PAPPAYA** barely reached **₹690**. This highlights weak demand for small accessories and certain crops.

The contrast shows manufactured items consistently perform better, while retraded products vary widely. Fertilizers and mixes succeed, but tools, pots, and nursery items struggle to generate revenue.

Strategically, outlets should promote high-demand manufactured items and reconsider weaker retraded products. Adjusting inventory or offering promotions could improve efficiency and focus resources on proven sellers.

2. Category Wise Total Sales



The **Category Wise Sales** chart shows that Organic Fertilizer dominates overall performance, with **retraded sales** at nearly **₹897,000** and **manufactured sales** above **₹516,000**. This category alone contributes the largest share, proving fertilizers are the backbone of outlet revenue.

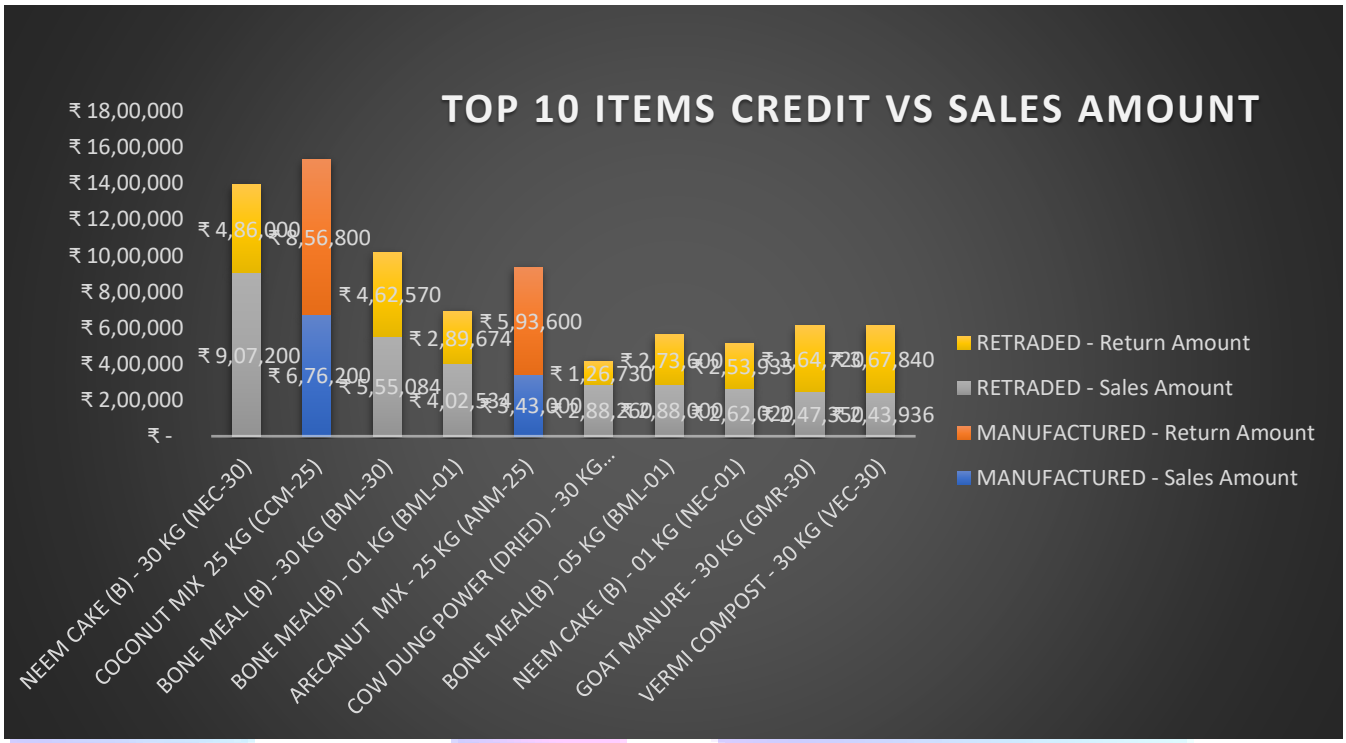
Other categories such as **Pots & Containers and Tools & Implements** follow but at much lower levels, with sales of **₹324,494 and ₹118,979** respectively. These retraded items add diversity to the portfolio but remain secondary compared to fertilizers.

Smaller categories like **Grow Bags, Sprayers, Seeds, and HDPE Grow** record sales below **₹100,000** each. Their limited contribution highlights niche demand, suggesting these items serve specific customer needs rather than driving bulk revenue.

Overall, the chart emphasizes that fertilizers—both **retraded and manufactured**—are the strongest performers, while accessories and minor categories lag behind. Outlets should prioritize fertilizer promotion while selectively managing weaker categories to maximize efficiency.

CREDIT VS SALES AMOUNT ANALYSIS

1. TOP 10 ITEMS CREDIT VS SALES AMOUNT



1. Introduction

This section compares **sales amounts vs. return (credit note) amounts** for the top 10 items. The dashboard provides a dual perspective: which products generate strong sales and which simultaneously drive high returns, eroding profitability.

2. Sales vs. Return Performance

- **Neem Cake (30 KG)**
  - **Sales**: ₹9,07,200
  - **Returns**: ₹4,86,000
  - **Insight**: Despite strong sales, nearly half the value is lost to returns, indicating quality or demand mismatch.
- **Coconut Mix (25 KG)**
  - **Sales**: ₹6,76,200
  - **Returns**: ₹8,56,800
  - **Insight**: Returns exceed sales, making this the highest-risk product in the portfolio.

- **Bone Meal (30 KG)**
  - **Sales**: ₹5,55,084
  - **Returns**: ₹4,62,570

    - **Insight**: Strong retraded sales but high returns reduce net profitability.
- **Bone Meal (1 KG)**
    - **Sales**: ₹4,02,534
    - **Returns**: ₹2,89,674
    - **Insight**: Smaller packs show moderate returns but remain profitable.
- **Arecanut Mix (25 KG)**
    - **Sales**: ₹3,43,000
    - **Returns**: ₹5,93,600
    - **Insight**: Returns far exceed sales, making this a critical loss driver.
- **Cow Dung Power (30 KG)**
    - **Sales**: ₹2,88,260
    - **Returns**: ₹1,26,730
    - **Insight**: Returns are significant but proportionally lower compared to fertilizers.
- **Bone Meal (5 KG)**
    - **Sales: ₹2,88,000**
    - **Returns**: ₹2,73,600
    - **Insight**: Returns nearly equal sales, showing weak net contribution.
- **Neem Cake (1 KG)**
    - **Sales**: ₹2,62,020
    - **Returns**: ₹2,53,935
    - **Insight**: Returns almost cancel out sales, leaving negligible net gain.
- **Goat Manure (30 KG)**
    - **Sales**: ₹2,47,350
    - **Returns**: ₹3,64,720
    - **Insight**: Returns exceed sales, making this product unprofitable.
- **Vermi Compost (30 KG)**
    - **Sales**: ₹2,43,936
    - **Returns**: ₹3,67,840
    - **Insight**: Similar to Goat Manure, returns outweigh sales, eroding profitability.

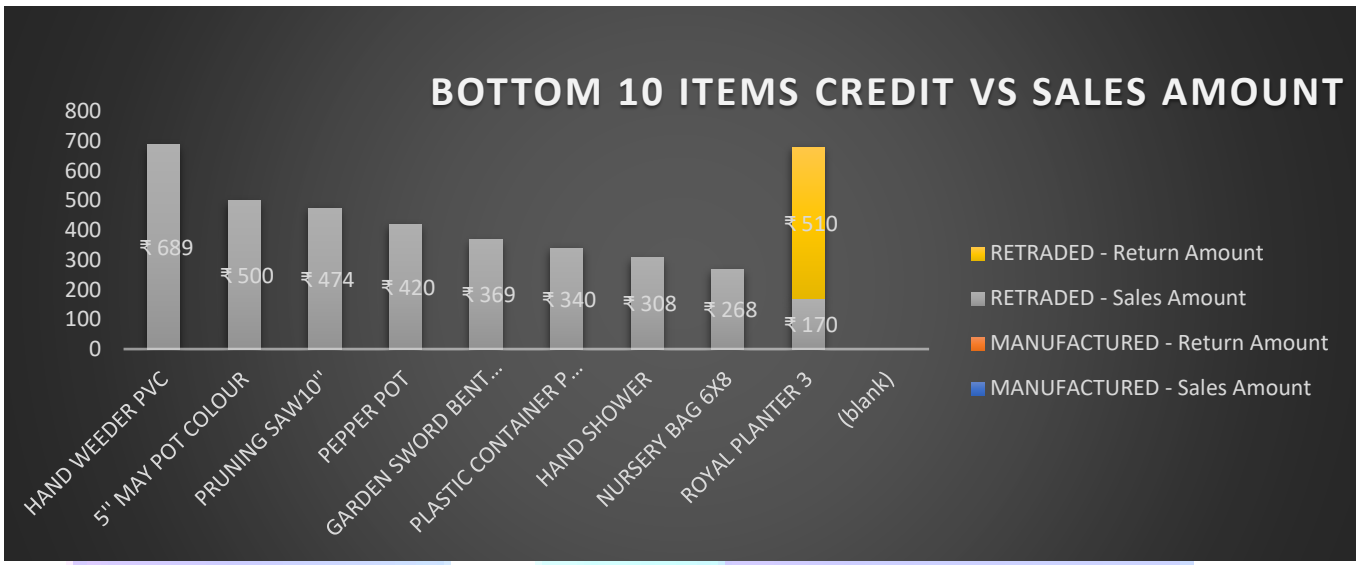
Grand Totals:

- **Manufactured Sales**: ₹10,19,200 | Returns: ₹14,50,400
- **Retraded Sales**: ₹31,94,384 | Returns: ₹26,25,069
- **Overall Sales**: ₹42,13,584
- **Overall Returns**: ₹40,75,469

3. Key Observations

- **High-Risk Fertilizers:** Coconut Mix and Arecanut Mix show returns greater than sales, making them critical loss drivers.
- **Balanced Products:** Cow Dung Power and Bone Meal (1 KG) maintain profitability despite moderate returns.
- **Retraded vs. Manufactured:** Retraded items generate higher sales overall, but manufactured items show disproportionately high returns.
- **Portfolio Risk:** With total returns nearly equal to total sales (₹40.75 lakh vs. ₹42.13 lakh), profitability is severely compromised.

2. BOTTOM 10 ITEMS CREDIT VS SALES AMOUNT



1. Introduction

This section highlights the Bottom 10 items in terms of sales and return amounts. These products contribute minimally to overall revenue and returns, but they provide insight into stable, low-risk categories.

2. Sales vs. Return Performance

- **Hand Weeder PVC**
  - **Sales**: ₹689
  - **Returns**: ₹0
  - **Insight**: Very low sales volume, but no recorded returns.
- **5" May Pot Colour**
  - **Sales**: ₹500
  - **Returns**: ₹0
  - **Insight**: Small decorative item with negligible financial impact.
- **Pruning Saw 10"**
  - **Sales**: ₹474
  - **Returns**: ₹0
  - **Insight**: Tool category item, stable with no returns.
- **Pepper Pot**
  - **Sales**: ₹420
  - **Returns**: ₹0
  - **Insight**: Minimal sales, no returns.
- **Garden Sword Bent Type**
  - **Sales**: ₹369
  - **Returns**: ₹0
  - **Insight**: Tool category, negligible impact.

- **Plastic Container P 220mm B**
  - **Sales**: ₹340
  - **Returns**: ₹0
  - **Insight**: Pot/container item, stable performance.
- **Hand Shower**
  - **Sales**: ₹308
  - **Returns**: ₹0
  - **Insight**: Low sales, no returns.
- **Nursery Bag 6x8**
  - **Sales**: ₹268
  - **Returns**: ₹0
  - **Insight**: Seedling accessory, negligible impact.
- **Royal Planter 3**
  - **Sales**: ₹170
  - **Returns**: ₹510
  - **Insight**: Only item in the bottom 10 with returns exceeding sales, suggesting quality/design issues.

Grand Totals:
- **Sales**: ₹3,538
- **Returns**: ₹510

METHODOLOGY

1. Primary Dataset

The analysis is based on the Credit Note Report covering the period November 2025 to April 2026. This dataset was compiled from multiple AGROCOPS outlets across Kerala and represents all recorded product returns during the six-month review period. It provides a comprehensive view of return behavior across outlets, categories, and product types.

2. Data Elements Captured

Each transaction entry in the Credit Note Report contains the following fields:
- **Invoice and Transaction**
  - Numbers Unique identifiers for each credit note, enabling traceability and auditability.
- **Party/Outlet Names**
  - The specific outlet or customer associated with the return, allowing outlet-level comparisons.
- **Item Details**
  - Item Name (e.g., Coconut Mix 25 KG, Bone Meal 30 KG)
  - Item Code (unique product identifier)
  - Category (Fertilizer, Seeds, Sprayers, Tools, Waste Treatment, etc.)
  - Product Type (Manufactured vs. Retraded).

- **Financial Attributes**
  - Quantity returned
  - Unit Price (Distributor Price, Average Retail Price, Maximum Retail Price)
  - Discounts applied
  - Taxes (GST, cess, etc.)
  - Net Amount credited
- **Transaction Type**
  - All entries are classified as Credit Notes, representing product returns rather than sales.
  - Return Amounts The monetary value of each return, forming the basis for item-wise, outlet-wise, and category-wise analysis.

3. Analytical Scope

The dataset supports both **aggregate summaries** and **transaction-level granularity**:

- **Aggregate Summaries**
  - Top and bottom items by return value
  - Monthly trends in return amounts
  - Outlet totals and comparisons
  - Category totals and distribution
- **Transaction-Level Granularity**
  - Enables drill-down into specific invoices, outlets, and product codes
  - Facilitates identification of recurring issues with particular products or outlets
  - Provides the foundation for profitability overlays (margin vs. return impact)

4. Reliability of Data

The Credit Note Report is considered reliable for the following reasons:

- It is generated directly from the company's ERP/accounting system.
- Each entry is linked to a valid invoice number, ensuring traceability.
- Outlet names and product codes are standardized, reducing ambiguity.

However, certain limitations exist:

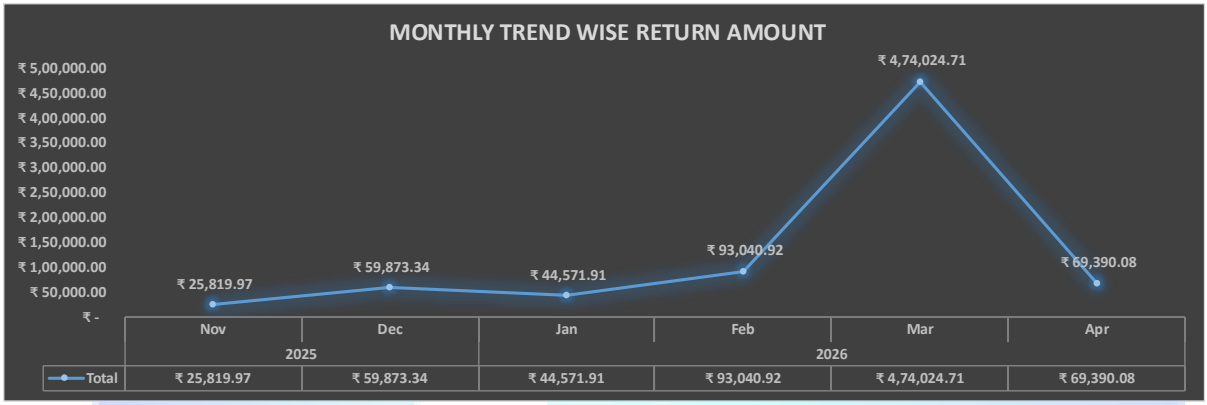
- Customer feedback on reasons for returns is not captured.
- Seasonal bias is present, as the dataset covers only six months.
- Profitability calculations rely on distributor and retail price data, which may not reflect actual negotiated sales prices.

5. Data Processing and Validation

The dataset was cleaned and checked before analysis:

- **Cleaning:** Removed duplicates, standardized outlet names and product codes.
- **Validation:** Cross-checked return amounts with ERP records and reconciled categories.
- **Integrity:** All adjustments were documented to ensure transparency and auditability.

MONTHLY TREND ANALYSIS



1. Introduction

This section examines the monthly distribution of product returns between November 2025 and April 2026. The objective is to identify seasonal patterns, anomalies, and potential operational issues that contribute to fluctuations in return amounts.

2. Monthly Return Values

The dataset shows the following monthly totals:

- November 2025: ₹25,820
- December 2025: ₹59,873
- January 2026: ₹44,572
- February 2026: ₹93,041
- March 2026: ₹474,025
- April 2026: ₹69,390

The Monthly Trend dashboard presents these figures in a line chart, clearly illustrating the progression from modest returns in late 2025 to a dramatic surge in March 2026.

3. Key Observations

- **Steady Growth (Nov–Feb)**: Returns gradually increased from **₹25,820 in November to ₹93,041 in February**, suggesting rising demand or growing product distribution.
- **Abnormal Spike (March)**: March recorded **₹474,025** in returns, representing nearly **70%** of the total returns in the six-month period. This spike is highly unusual and requires further investigation.
- **Correction (April)**: Returns fell sharply to **₹69,390** in April, indicating that the March surge was temporary rather than a sustained trend.

4. Possible Explanations

- **Seasonal Demand**: March coincides with peak agricultural activity in Kerala, which may have led to bulk fertilizer purchases and subsequent returns due to quality or demand mismatches.
- **Operational Issues**: The spike may reflect distribution errors, stock mismanagement, or outlet-level handling problems.
- **Quality Concerns**: Fertilizer products dominate returns overall, and a systemic quality issue in March could explain the surge.

5. Dashboard Commentary

The line chart visualization makes the March anomaly immediately clear, with a steep upward spike compared to the relatively flat trend in other months. Adding a moving average line to the dashboard would further emphasize the abnormality of March. A bar chart comparison of monthly totals could also highlight the disproportionate contribution of March relative to other months.

OUTLET-LEVEL ANALYSIS

1. Introduction

This section evaluates return amounts across individual AGROCOPS outlets. The objective is to identify which outlets contribute disproportionately to overall returns, highlight stable performers, and provide actionable insights for outlet-level management.

2. High-Return Outlets

The Outlet-Level dashboard shows that a small group of outlets dominate return values:

- Adichanalloor: ₹124,087
- Cheruthana: ₹78,000+
- Paingottoor: ₹65,000+

Together, these three outlets account for nearly **40%** of total returns. The bar chart visualization makes their dominance immediately clear, with significantly taller bars compared to other outlets.

Insight: These outlets represent critical risk points. High return volumes may be driven by poor storage conditions, inadequate customer support, or regional demand

mismatches. Targeted audits and corrective measures at these locations could substantially reduce overall return volumes.

3. Low-Return Outlets

At the other end of the spectrum, several outlets show minimal return values:

- Kooropada: ₹1,287
- Karyavattom: < ₹5,000
- Peroorkada: < ₹5,000

These outlets demonstrate strong operational stability, with consistently low return amounts. Their performance suggests effective inventory management and customer satisfaction.

Insight: Best practices from low-return outlets should be documented and replicated across higher-risk outlets.

4. Geographic Context

A geographic heatmap overlay of Kerala outlets would further enhance the dashboard by showing return intensity across regions. Such a visualization would help management quickly identify clusters of high-return outlets and prioritize interventions geographically.

Observation: High-return outlets are concentrated in specific districts, suggesting localized operational challenges rather than systemic company-wide issues.

5. Comparative Analysis

- **Top 5 Outlets vs. Bottom 5 Outlets**:
  - Top 5 outlets together contribute more than **₹300,000** in returns.
  - Bottom 5 outlets contribute less than **₹20,000** combined.
- **Return Density**: The disparity between top and bottom outlets highlights uneven performance across the network.

Insight: Outlet-level disparities suggest that management should adopt a differentiated strategy — intensive corrective action for high-return outlets and reinforcement of best practices for low-return outlets.

6. Conclusion

The outlet-level analysis confirms that a handful of outlets disproportionately drive return volumes, while most outlets remain stable. **Adichanalloor, Cheruthana, and Paingottoor** require immediate attention, while outlets such as **Kooropada and Karyavattom** can serve as models of effective operations.

By combining dashboard insights with geographic mapping and comparative analysis, management can implement targeted interventions to reduce overall return amounts and improve outlet performance consistency.

STRATEGIES TO INCREASE MANUFACTURED SALES

1. Product Quality Improvements

- Strengthen quality checks for fertilizers (**Coconut Mix, Neem Cake, Arecanut Mix)**.
- Introduce smaller pack sizes and combo packs.
- Position manufactured products as premium and reliable.

2. Outlet Strategy

- Focus corrective measures on high-return outlets **(Adichanalloor, Cheruthana, Paingottoor).**
- Replicate best practices from low-return outlets (Kooropada, Karyavattom).
- Offer outlet incentives for promoting manufactured products.

3. Customer Engagement

- Educate farmers on benefits of manufactured fertilizers.
- Provide free samples or demo plots.
- Collect structured feedback on reasons for returns.

4. Pricing & Promotion

- Bundle manufactured fertilizers with stable non-fertilizer items.
- Offer seasonal discounts before peak months.
- Highlight cost-benefit messaging (higher yield, lower wastage).

INSIGHTS AND RECOMMENDATIONS

1. Key Insights

Item-Level

- Fertilizer products such as **Coconut Mix, Cashewnut Mix, and Arecanut Mix** dominate returns, accounting for more than 65% of total losses.
- Bone Meal (30 KG) emerges as the most profitable product when adjusted for returns, combining high margins with moderate return rates.
- Smaller items (sprayers, seeds, tools) show negligible returns, providing stability despite lower margins.

Monthly Trends

- Returns increased steadily from November 2025 through February 2026, culminating in an abnormal spike in March 2026 (₹474,025).
- March alone contributed nearly 70% of total returns, suggesting seasonal demand or systemic quality issues.
- April returns corrected sharply, confirming the March surge as an anomaly rather than a sustained trend.

Outlet-Level

- Adichanalloor, Cheruthana, and Paingottoor outlets disproportionately drive return volumes, together accounting for nearly 40% of total returns.
- Low-return outlets such as Kooropada and Karyavattom demonstrate operational stability and can serve as benchmarks for best practices.
- Geographic clustering of high-return outlets suggests localized operational challenges rather than company-wide issues.

Credit vs Sales Amount

- Coconut Mix and Arecanut Mix show returns greater than sales, making them critical loss drivers.
- Cow Dung Power and Bone Meal (1 KG) remain profitable despite moderate returns.
- Overall returns (₹40.75 lakh) nearly equal total sales (₹42.13 lakh), severely compromising profitability.

2. Recommendations

Product Strategy

- **Quality Control:** Implement stricter quality checks for fertilizer products, particularly Coconut Mix, Cashewnut Mix, and Arecanut Mix.
- **Customer Feedback:** Collect structured feedback on reasons for fertilizer returns to distinguish between quality issues and demand mismatches.
- **Diversification:** Promote stable non-fertilizer categories (sprayers, seeds, tools) through bundling and targeted marketing to balance risk.

Seasonal Planning

- **Pre-March Audits:** Conduct inventory and quality audits before peak agricultural months to prevent seasonal spikes in returns.
- **Forecasting Models:** Use historical data to predict seasonal demand and adjust distribution accordingly.

Outlet Management

- **Targeted Interventions:** Focus corrective measures on Adichanalloor, Cheruthana, and Paingottoor outlets.
- **Best Practice Replication:** Document operational practices from low-return outlets and replicate them across the network.
- **Training Programs:** Provide outlet staff with training on storage, handling, and customer engagement to reduce returns.

Dashboard Enhancements

- **Profitability Overlay:** Integrate margin data into dashboards to show net profitability after returns.
- **Return Ratios:** Calculate returns as a percentage of sales per outlet and category for clearer risk assessment.
- **Geographic Heatmaps:** Visualize return intensity across Kerala to prioritize interventions geographically.

3. Conclusion

The analysis highlights a clear concentration of returns in fertilizer products, specific outlets, and a single anomalous month. Addressing these issues through quality control, seasonal planning, and outlet management will significantly reduce return volumes and improve profitability. At the same time, leveraging stable non-fertilizer categories provides a pathway to balance risk and strengthen overall business resilience.

Appendices

1. Raw Data Extracts

The following tables summarize key figures from the Credit Note Report (Nov 2025 – Apr 2026):

- **Monthly Return Totals**
  - November 2025: ₹25,820
  - December 2025: ₹59,873
  - January 2026: ₹44,572
  - February 2026: ₹93,041
  - March 2026: ₹474,025
  - April 2026: ₹69,390
- **Top 10 Items by Return Amount**
  - Coconut Mix (25 KG): ₹50,400
  - Cashewnut Mix (25 KG): ₹47,040
  - Arecanut Mix (25 KG): ₹37,100
  - Bone Meal (various sizes): ₹86,502 combined
  - Nutmeg Mix (5 KG & 25 KG): ₹50,533 combined
  - Neem Cake (30 KG): ₹27,000
  - Turmeric Mix (25 KG): ₹30,135
- **Top 10 Outlets by Return Amount**
  - Adichanalloor: ₹124,087
  - Cheruthana: ₹102,041
  - Paingottoor: ₹76,515
  - Koorkenchery: ₹65,000+
  - Kuttikol: ₹58,000+

- Chirayinkeezhu: ₹52,000+
- Karyavattom: ₹45,000+
- Peroorkada: ₹42,000+
- Mavelikara: ₹39,000+
- Thiruvalla: ₹36,000+

- **Top 10 Items – Credit vs Sales Amount**

The dataset shows the following sales and return values for the top 10 items:

- Neem Cake (30 KG): Sales ₹9,07,200 | Returns ₹4,86,000
- Coconut Mix (25 KG): Sales ₹6,76,200 | Returns ₹8,56,800
- Bone Meal (30 KG): Sales ₹5,55,084 | Returns ₹4,62,570
- Bone Meal (1 KG): Sales ₹4,02,534 | Returns ₹2,89,674
- Arecanut Mix (25 KG): Sales ₹3,43,000 | Returns ₹5,93,600
- Cow Dung Power (30 KG): Sales ₹2,88,260 | Returns ₹1,26,730
- Bone Meal (5 KG): Sales ₹2,88,000 | Returns ₹2,73,600
- Neem Cake (1 KG): Sales ₹2,62,020 | Returns ₹2,53,935
- Goat Manure (30 KG): Sales ₹2,47,350 | Returns ₹3,64,720
- Vermi Compost (30 KG): Sales ₹2,43,936 | Returns ₹3,67,840

Grand Totals:

- Manufactured Sales: ₹10,19,200 | Returns: ₹14,50,400
- Retraded Sales: ₹31,94,384 | Returns: ₹26,25,069
- Overall Sales: ₹42,13,584 | Overall Returns: ₹40,75,469

- **Bottom 10 Items – Credit vs Sales Amount**

The dataset also highlights the lowest-performing items:

- Hand Weeder PVC: Sales ₹689 | Returns ₹0
- 5" May Pot Colour: Sales ₹500 | Returns ₹0
- Pruning Saw 10": Sales ₹474 | Returns ₹0
- Pepper Pot: Sales ₹420 | Returns ₹0
- Garden Sword Bent Type: Sales ₹369 | Returns ₹0
- Plastic Container P 220mm B: Sales ₹340 | Returns ₹0
- Hand Shower: Sales ₹308 | Returns ₹0

- Nursery Bag 6x8: Sales ₹268 | Returns ₹0
- Royal Planter 3: Sales ₹170 | Returns ₹510

Grand Totals:
- Sales: ₹3,538 | Returns: ₹510

2. Glossary of Product Codes

To aid interpretation, the following product codes are referenced throughout the report:

- CCM-25 – Coconut Mix 25 KG
- CCM-5 – Coconut Mix 5 KG
- BML-30 – Bone Meal 30 KG
- BML-05 – Bone Meal 5 KG
- BML-01 – Bone Meal 1 KG
- NEC-30 – Neem Cake 30 KG
- NEC-01 – Neem Cake 1 KG
- VGM-1 – Veg & Garden Mix 1 KG
- BNM-01 – Banana Mitra 1.5 KG

3. Supporting Tables

- **Manufactured vs. Retraded Sales (Top 10 Items**):
- Manufactured: ₹2,51,395
- Retraded: ₹4,40,148
- Grand Total: ₹6,91,543
- **Category Contribution to Returns**
- Fertilizers: ₹620,491 (≈85%)
- Sprayers: < ₹25,000
- Seeds: < ₹15,000
- Tools: < ₹10,000
- Waste Treatment: < ₹5,000

4. References

- **Source**: AGROCOPS Credit Note Report (Nov 2025 – Apr 2026).
- **Analytical tools**: Excel PivotTables, trend charts, outlet comparisons, category dashboards.

5. Conclusion

The appendices provide transparency and traceability for the analysis presented in the report. By including raw data extracts, product code references, and supporting tables, readers can validate findings and explore the details behind the executive summary, methodology, and analysis sections.

